

GROUP WORK PROJECT MScFE 600: FINANCIAL DATA

Group Number:11269

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Rhushabh Vaghela	Ahmedabad, India	rhushabhvaghela@gmail.com	
Sheharyar Hassan Mirza	Islamabad, Pakistan	sheharyar95@live.com	
Iris Yang	Los Angeles, CA, USA	irisyang886@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	Rhushabh Vaghela
Team member 2	Sheharyar Hassan Mirza
Team member 3	Iris Yang

Group Number: 11269

Table of Contents

Part 1 — Assessing Models with Alternative Data

Q1. Data Understanding

Q2. Security Understanding (ECH case)

Q3. Methodology Understanding

Q4. Feature Understanding

Q5. Optimization Understanding

Step 1 — The Financial Problem

Step 2 — Application

Step 3 — Replication of Methodology (IVV)

Part 2 — Evaluating One Particular Type of Alternative Data

1. Sources of data

2. Types of data used

3. Data quality considerations missing/outliers.

4. Ethical issues

5. Python demonstrations

6. Exploratory findings (summary) — distributional notes and class imbalance.

7. Short literature connections — credit/alt-data performance & risks (select citations).

References — datasets and scholarly sources.

Group Number: 11269

Writing Report

Part 1 - Assessing Models with Alternative Data

In this section we have answered the following questions after assessing the “An Intelligent Approach for Predicting Stock Market Movements in Emerging Markets Using Optimized Technical Indicators and Neural Networks” by Sagaceta Mejia et al research paper.

Q1. Data Understanding:

Types of Data and Indicators: The research utilizes daily Open, High, Low, Close, and Volume (OHLCV) data for three Exchange-Traded Funds (ETFs): iShares MSCI Chile (ECH), iShares MSCI Brazil (EWZ), and iShares Core S&P 500 (IVV). The authors use the Pandas Technical Analysis library to add 210 more technical indicators to these basic data points. This makes a complete dataset with about 216 features per day. There are many types of indicators, such as momentum, volatility, trend, and volume. The data is cleaned by removing rows that don't have full lookback windows for calculating indicators and scaling all features to a common 0-1 scale.

Importance of Indicators: Indicators are important because they turn raw, often messy price and volume data into market states that people can understand. They help find things like trend direction, overbought/oversold levels, and price-volume alignment, which give a machine learning model a clearer signal to learn from. The main finding of the study is that the Selected(5) set, which contains features that at least five different selection methods agree on, gives better accuracy while using only about 5% of the total features. This shows that a small, useful set of indicators works better than just using all the indicators.

Group Number: 11269

Q2. Security Understanding:

For our analysis, we selected the **iShares MSCI Chile ETF (ECH)**, one of the two emerging market funds examined in the paper by Sagaceta-Mejía et al. (2024). This fund serves as an excellent case study for the paper's objective of predicting market movements. ECH aims to track the MSCI Chile IMI 25/50 Index, offering investors a direct but passive investment vehicle into the performance of the Chilean equity market. As a single-country emerging market fund, it encapsulates the distinct volatility and growth characteristics that differentiate it from developed market ETFs like the IVV.

Reviewing the fund composition described, during the period under study (2009-2020) Chilean economic drivers were directly reflected. The top sector exposures are:

Financials (21.53%): It consists of the majority of allocated funds, typical of a developing economy building its capital markets.

Materials (21.28%): This stands to be the most defining characteristic of the fund as it is heavily invested in Chile's gigantic copper mining industry. This renders ECH highly sensitive to global commodity cycles and industrial demand, particularly from China.

Utilities (18.73%): These regulated firms provide a defensive counterbalance to highly cyclical materials, representing the stable, domestic side of the Chilean economy.

Performance Analysis and Volatility Profile (2009–2020)

The performance of ECH during the analysis period underscores the high-risk, high-return nature of emerging market investing. The fund's opening price ranged from a low of \$29.30 to a high of \$80.25, demonstrating significant price swings.

Compared to the IVV (S&P 500), ECH's price history was far more volatile and lacked a consistent long-term trend. This volatility is not random; it is a function of real-world risk factors, including:

Group Number: 11269

Commodity Price Fluctuation: The fund's value was heavily influenced by the global price of copper.

Currency Risk: Fluctuations between the Chilean Peso (CLP) and the U.S. Dollar (USD) directly impact returns for foreign investors.

Political and Social Factors: The period included significant events like the 2010 earthquake and periods of social unrest, which introduced market uncertainty.

Rationale and Alternatives of Classification

This is especially true since the classification adopted in this paper (the prediction "up" or "down") has many reasons to be favored for action above trying to predict an exact price, which could be extremely volatile due to the presence of market noise and outliers.

There could have been other frameworks for classification:

Three-State Model (Up/Flat/Down): Another option could be the introduction of a neutral band (e.g. +/- 0.5% daily change) to filter out noise-driven price movements of minor contours. Thus, any movement occurring within this band would be "flat" and it would not tend to be captured by the model, which would instead concentrate on much bigger tradeable signals.

Trend-Based Regime: Another method could be to classify each day in relation to a short-term moving average (e.g. the 10-day MA). If the price crosses above the average, the label could be "1" and if below, the label could be "-1". In this way, predictability would be moved away from noise on single days, and will instead be directed toward capturing broader shifts in market trend or regime.

Group Number: 11269

Q3. Methodology Understanding

Restructured Outline:

Section 2: Data

2.1 Stocks Analyzed

2.2 Technical Indicator Generation

2.3 Data Preparation

2.3.1 Class Assignment

2.3.2 Data Normalization

2.3.3 Data Cleaning

Section 3: Methodology

3.1 Feature Selection Techniques

3.1.1 *Descriptive Statistics* (Low Variance, Pearson's Correlation, MAD, Dispersion Ratio)

3.1.2 *Statistical Models* (Chi-Squared, LASSO, Tree-based Selection, Principal Feature Analysis)

3.2 *Predictive Model*: Multilayer Perceptron (MLP)

3.3 *Model Evaluation*: 10-Fold Cross-Validation

Descriptive Statistics vs. Models:

Descriptive statistics provide in-depth summaries and descriptions for relationships the data shares. They are used to filter relevant characteristics based on their attributes. Pearson's correlation measures the linear relationship between two variables without forming predictive formulas.

Models are those algorithms that learn predictions from data. In this sense, LASSO is a model; it is a regularized version of regression that builds a predictive equation and uses the penalty on the weights for simultaneous selection of features.

Group Number: 11269

Optimization Process: An optimization process is one in which the given optimization is that of the subset of technical indicators against the parameters of these indicators. Following is the pipeline that was used in the research -

- A universe of ~**216 indicators** is generated.
- Use of **eight feature selection** techniques to rank the indicators according to their importance.
- Forming consensus-based subsets called **Selected(n)**, containing only those indicators which appear in the top quartile of at least n selection methods.
- Evaluate each subset with an MLP model using 10-fold cross-validation to find the combination that maximizes accuracy with minimal complexity.

This optimization is crucial for the neural network because it reduces overfitting, improves computational efficiency by lowering the number of inputs, and enhances predictive accuracy by removing redundant and noisy features.

Q4. Feature Understanding

Feature: A feature is any input column taken by a model to predict outcomes. These include raw OHLCVs and derived from 210 other technical indicators.

Method: The method is the way of features selection and analysis. For example MAD (Mean Absolute Difference) is a method for ranking features in accordance with their discrimination ability.

Model: A model is a forecasting engine that takes features and gives output from them. The model is the Multilayer Perceptron that takes features as input in order to predict market direction.

Feature Categories: The Pandas-TA library collects indicators in ten categories. Candles, Cycles, Momentum, Overlap, Performance, Statistics, Trend, Utility, Volatility, and Volume.

Group Number: 11269

Optimization Process: Instead of customizing every parameter of each indicator originating from the study, it optimizes the set of features fed into the model. This is done by aggregating rankings from several methods of selection and keeping only the consistently high-scoring indicators across the various techniques. The resulting **Selected(5)** set is small yet diverse in that it encompasses momentum, trend, volatility, and volume dynamics. This removes redundant structures so that model training becomes faster and, much more importantly, out-of-sample accuracy increases due to the creation of a more robust and more generalizable model.

Q5. Optimization Understanding

1. Cross-Validation:

- **Cross-validation** can describe how well the results of a model are expected to generalize to an independent dataset. This technique systematically partitions data into training and testing data, thus providing assurance that the performance metrics are credible and not merely a product of a fortunate single train-test split.
- **K-Fold Cross-Validation:** This is a special type in which all k folds, or groups, are created from the data. The model is trained and tested k times; each time, one fold is used for testing and the remaining k-1 folds are used for training. Final performance results from the average for k trials. In this paper, it uses k=10.

2. Jaccard Distance and Comparison:

- **Jaccard Distance:** It compares two sets in terms of the difference between them. It is defined in terms of the formula $J(A, B) = |A \cap B| / |A \cup B|$, which computes the fraction of elements which are not common to both sets out of all unique elements. In this paper, it is used to determine how far apart important feature sets are for emerging market ETFs (ECH, EWZ) regarding the developed market ETF (IVV). It was concluded that emerging market ETFs have more indicators in common with each other than either has with the developed market ETF.

Group Number: 11269

- **Comparison to Other Metrics:**

A. Comparison with Euclidean Distance: The Jaccard distance is typically used for the calculation of dissimilarities in sets, or in binary data which have indications of either being in the set or not. On the other hand, Euclidean distance is defined for vectors whose components can take values in a continuous range in such a way that they identify points in the monitor space of random dimension. While Jaccard is an excellent way to compare the feature lists' contents, the concept of distance with Euclidean notion does not apply here.

B. Comparison with Cosine Distance: Cosine distance speaks more about the difference in orientation (angle) between two numerical vectors, disregarding their magnitude. This method of measuring distance is common in mathematical text analysis, but establishing distances using Jaccard gives a more straightforward answer to the question of how many unshared items exist between two finite lists of features.

Optimal Solution: The optimal solution, according to the paper, is the least set size of features that maximizes the model's predictive performance in terms of cross-validation accuracy. The Selected(5) feature subset is identified as such because it strikes an "accuracy-efficiency sweet spot," achieving higher accuracy than the full feature set while garbing itself in substantially greater computational efficiency.

Step 1. The Financial Problem

The financial problem being tackled is constructing a specific system for the prediction of near-term direction behavior of exchange-traded funds or ETFs in the stock market. Such a model will serve as the main determinant of decisions by investors on two main fronts of financial activity:

Market Timing: It attempts to recognize moments where entering and exiting positions may augment price profits or minimize losses.

Group Number: 11269

Risk Management: Change investment strategy according to forecasted changes in the market to minimize risks associated with volatility. The primary challenge is filtering through the massive number of available indicators to extract the few that are truly predictive, rather than relying on an inefficient "throw everything in" approach to do this.

Predicting for Emerging Markets Vs Developed Markets: The actual nature of prediction of movements in emerging markets is very different from that of developed markets; these differences stem from variant traits of the economies. It is indisputable that emerging economies tend to grow fast while being afflicted by **greater volatility and less liquidity** in their equity markets.

All this is indeed pertinent to the construction of such a model, as the very factors causing this kind of movement in the market behave differently from these two market types. To provide evidence of this, the study showed how the best predictive indicator set (**Selected(5)**) differed for one group of emerging market ETFs, (ECH and EWZ) than that for the developed market ETF (IVV). Jaccard distance metric proved the fact that the feature sets for the two emerging market ETFs were much closer to each other than any could be to the developed market ETF. This means, therefore, that a "one-size-fits-all" predictive model is likely to be suboptimal and that successful design must be aligned to a specific market structure.

Step 2. Application

Results from the study yield important implications for financial modeling -

- **Feature Selection:** Most significant are the results supporting the strong necessity for intelligent feature selection. A small number of well-chosen indicators (approximately 5% of the number tested) increased prediction accuracy approximately 2% over the accuracy afforded by using the total feature set.

Group Number: 11269

- **Gains in Efficiency Are Enormous:** The gains in using such a small, optimized feature set were a quantifiable efficiency, where the average training time of the model was reduced by 84.68%.
- **No Universal Predictors:** A market-dependent situation exists with respect to the greatest set of features. The features found to best predict in emerging markets like Chile and Brazil are different from those which are most predictive in the developed U.S. market.
- **The "Sweet Spot" for Reduction:** There is an optimal level of feature reduction. While reducing features to the Selected(5) set was good, going further down to the Selected(6) set caused a drastic drop in accuracy, establishing the need for some core level of diverse information for a robust prediction.

Specific Useful Features The most useful features were those identified in the (Selected(5)) sets, which represent a consensus across multiple selection methods. While the exact combination varied for each ETF, several indicators proved to be broadly useful:

- **Common to All Three ETFs:** The features BBP (Bollinger Band Percent), BOP (Balance of Power), DEC (Decreasing), INC (Increasing), and J (from the KDJ indicator) were found in all three Selected(5) sets, suggesting they are robust predictors across different market types.
- **Common to Emerging Markets:** The features AOBV (Archer's On Balance Volume), CTI (Correlation Trend Indicator), and EBSW (Even Better SineWave) were selected for both ECH and EWZ, indicating their specific relevance to emerging markets.

Group Number: 11269

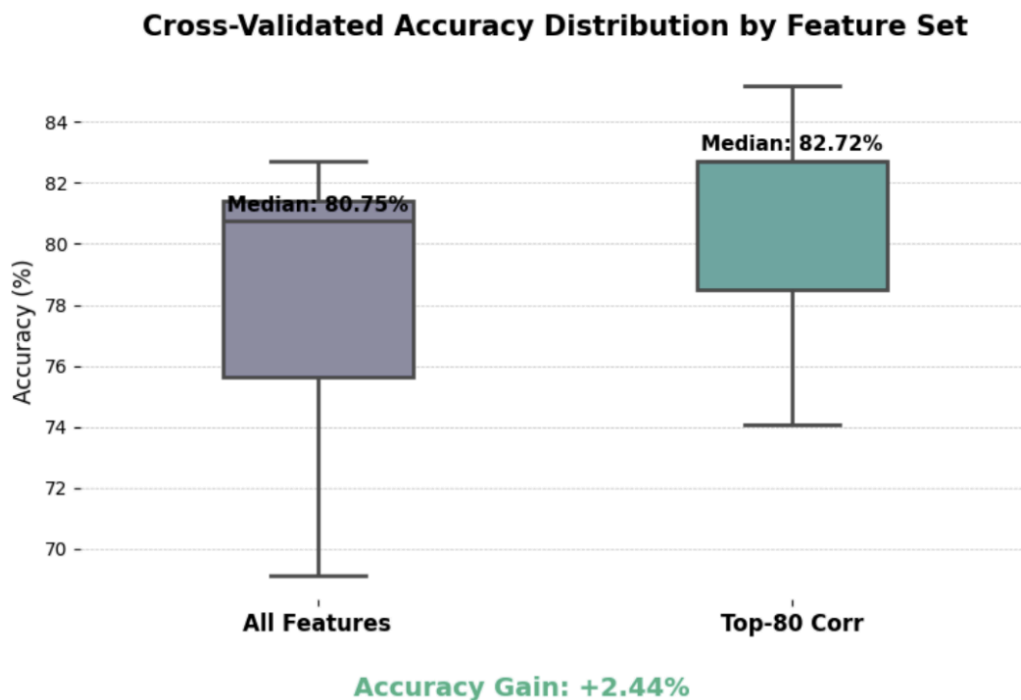
Step 3: Replication of Methodology

To put the main thesis of the paper to the test, a feature selection procedure was replicated. This step was performed to substantiate that from the entire, non-discriminated family of features, a selection of technical indicators, if optimized, would lend itself to the construction of a more accurate predictive model.

Methodology

The replication study was conducted for the iShares Core S&P 500 ETF (IVV), using daily data from January 1, 2019, to January 1, 2023. This process produced, as per the article, an exhaustive universe of technical indicators, extending beyond 200 from the base OHLCV data.

In this regard, feature selection was performed using the Pearson correlation coefficient as the ranking measure, for which the selection laid its basis. The absolute value of correlation for each of the indicators with the target binary variable, which was designed to distinguish in which direction the price shall go the next day, was computed. To gain the performance evaluation, the Logistic Regression model was tested and validated with the 10-fold stratified cross-validation method. The model was iterated by increasing the number of top-ranked features (from 1 to 100) each time to find the optimal size for the feature set.



Group Number: 11269**Results and Interpretation**

First of all, the replication was able to corroborate the findings of the paper. The peak model performance was observed with the top 80 most correlated features.

The model using Top-80 optimized feature set achieved a median cross-validated accuracy of 82.72%.

All 332 features provided a lower median accuracy of 80.75%, as compared to the Top-80 optimized set.

Intelligent feature selection therefore provides an accuracy gain of +2.44%. This intuitively strengthens the proposition that including irrelevant or noisy indicators hampers the predictive power of the model. Targeting and isolating the subset of most informative indicators allows the model to be not just computationally more efficiently trained, but also less overfitted and more accurate. The accompanying charts in the code appendix visually validate the results with a line plot showing model accuracy peaked at $k=80$ and a box plot presenting the enhanced performance distribution of the optimized model. This outcome becomes evidence in favor of the core argument of the paper: a disciplined feature selection process is a critical step in building effective predictive models for financial markets.

Group Number: **11269**

Part 2 - Evaluating One Particular Type of Alternative Data

In current times, credit data is the most valuable source for financial analysis. The data is a reflection of credit behavior of individuals as well as institutions and governmental bodies. It also serves as a key indicator of financial and economic stability. All policy makers including banks and financial institutions primarily depend on credit data for understanding of lending patterns, identifying risks and designing credit policies.

We used credit data in our report as a subcategory of analysis data. The discussion involves its sources, types, quality issues, and ethical considerations, followed by a Python-based demonstration of how such data can be imported and structured for analysis. Our objective is to show how credit data can be transformed from raw financial records into actionable economic information.

1. Sources of Data

Credit data originates primarily from financial institutions that record borrower and loan activity. These include commercial banks, credit card companies, microfinance institutions, and national credit registries.

Credit data can be accessed in various formats (CSV, Excel, API feeds) depending on the institution. Public data is often aggregated, while private credit bureaus provide individual-level datasets under strict confidentiality.

We used the following data sources for our analysis.

Source	Data Set	Access/Type	Strengths
German Credit Dataset	UCI Machine Learning Repository	German Machine Learning/CSV	Behavioral insights
CFPB	Consumer Complaints data	CFPB site/CSV	Demographic data
Kaggle	Kaggle datasets (Credit Card Fraud Detection)	Kaggle Datasets/CSV	Used for modelling

Group Number: 11269

2. Types of Data

Our German Credit Dataset data fell into quantitative and qualitative categories. Our analysis data was stored in structured, tabular format.

a. Quantitative Data

- Credit Risk
- A1 Status of existing checking account
- A7 Employment

b. Qualitative Data

- Age
- A4 Purpose
- A3 Credit history
- A14 other instalment plans

c. Alternative and Derived Data

The other data source we used was alternative and derived data based on consumer complaints for credit. We defined parameters for usage of data and did analysis of base complaints maximum number being credit complaints

3. Quality of Data

The reliability of analysis based on credit data depends heavily on data quality including:

1. Accuracy:

Errors may occur during manual entry or due to inconsistent data synchronization between financial institutions.

2. Completeness:

Not all lenders report to credit registries. In developing economies, informal lenders and microfinance providers may be excluded.

3. Consistency:

Differences in definitions—for example, how “default” or “delinquency” is classified—can distort cross-institution or cross-country comparisons.

4. Timeliness:

Credit data should be updated regularly; otherwise, outdated records can misrepresent borrower risk.

5. Bias:

Certain groups (e.g., women, rural borrowers, low-income individuals) may be underrepresented, leading to biased conclusions in credit scoring or inclusion policies.

6. Missing and Outlier Values:

Missing credit scores or extreme loan values can affect model training and must be handled through imputation or normalization.

Our datasets had no accuracy or completeness issues because these were properly formatted and received from recognized sources.

Group Number: 11269

4. Ethical Issues

Credit data may contain personally identifiable and sensitive financial information. Ethical handling of this data is crucial to prevent misuse and maintain public trust. The Kaggle dataset for credit card fraud has personally identified information encrypted in PCA driven columns V1, V2 etc.

1. Data Protection and Privacy:

Regulations e.g. General Data Protection Regulation (GDPR) in Europe and California Consumer Privacy Act (CCPA) in the U.S. require informed consent, secure storage, and limited data sharing.

2. Transparency:

Individuals should have the right to know how their data are collected, processed, and used in credit scoring or lending decisions.

3. Fairness and Non-Discrimination:

Machine learning models trained on biased data may unfairly deny credit to certain groups. Fairness auditing and bias mitigation must be applied.

4. Security:

Encryption, anonymization, and restricted access are mandatory to protect data from breaches and identity theft.

5. Purpose Limitation:

Credit data collected for financial purposes should not be reused for unrelated commercial or surveillance activities.

Ethical credit data management is not only a legal requirement but also a pivotal element of responsible data science practices.

5. Python Code to Import and Structure into Useful Data Structures

For German Credit Data

```
import pandas as pd
```

```
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/german.data"
```

```
columns = ['Status', 'Duration', 'CreditHistory', 'Purpose', 'CreditAmount', 'Savings',
```

```
'Employment',
```

```
        'InstallmentRate', 'PersonalStatusSex', 'Debtors', 'ResidenceSince', 'Property', 'Age',
```

```
        'OtherInstallmentPlans', 'Housing', 'ExistingCredits', 'Job', 'LiablePeople', 'Telephone',
```

```
        'ForeignWorker', 'CreditRisk']
```

```
df = pd.read_csv(url, sep=' ', header=None, names=columns)
```

```
print(df.head())
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Distribution of credit amount
```

```
sns.histplot(df['CreditAmount'], bins=30)
```

Group Number: 11269

```
plt.title("Distribution of Credit Amounts")
plt.xlabel("Credit Amount")
plt.ylabel("Frequency")
plt.show()

# Credit risk vs age
sns.boxplot(x='CreditRisk', y='Age', data=df)
plt.title("Age Distribution by Credit Risk")
```

For Consumer Complaints

```
import pandas as pd
import matplotlib.pyplot as plt

file_path = "complaints-2025-10-17_08_43.csv"

# Load a subset of columns for faster exploration
cols = [
    "Date received", "Product", "Issue", "Company",
    "State", "Submitted via", "Company response to consumer", "Timely response?"
]

complaints = pd.read_csv(file_path, usecols=cols, low_memory=False)

# Complaints per month
complaints["Month"] = complaints["Date received"].dt.to_period("M")
monthly = complaints.groupby("Month").size()

monthly.plot(title="Consumer Complaints Over Time")
plt.ylabel("Number of Complaints")
plt.show()

top_companies = complaints["Company"].value_counts().head(10)
top_companies.plot(kind="bar", title="Top 10 Companies by Complaint Count")
plt.ylabel("Complaints")
plt.show()
```

For Kaggle Credit Card Fraud

```
import pandas as pd

# Load CSV
data = pd.read_csv("creditcard.csv")

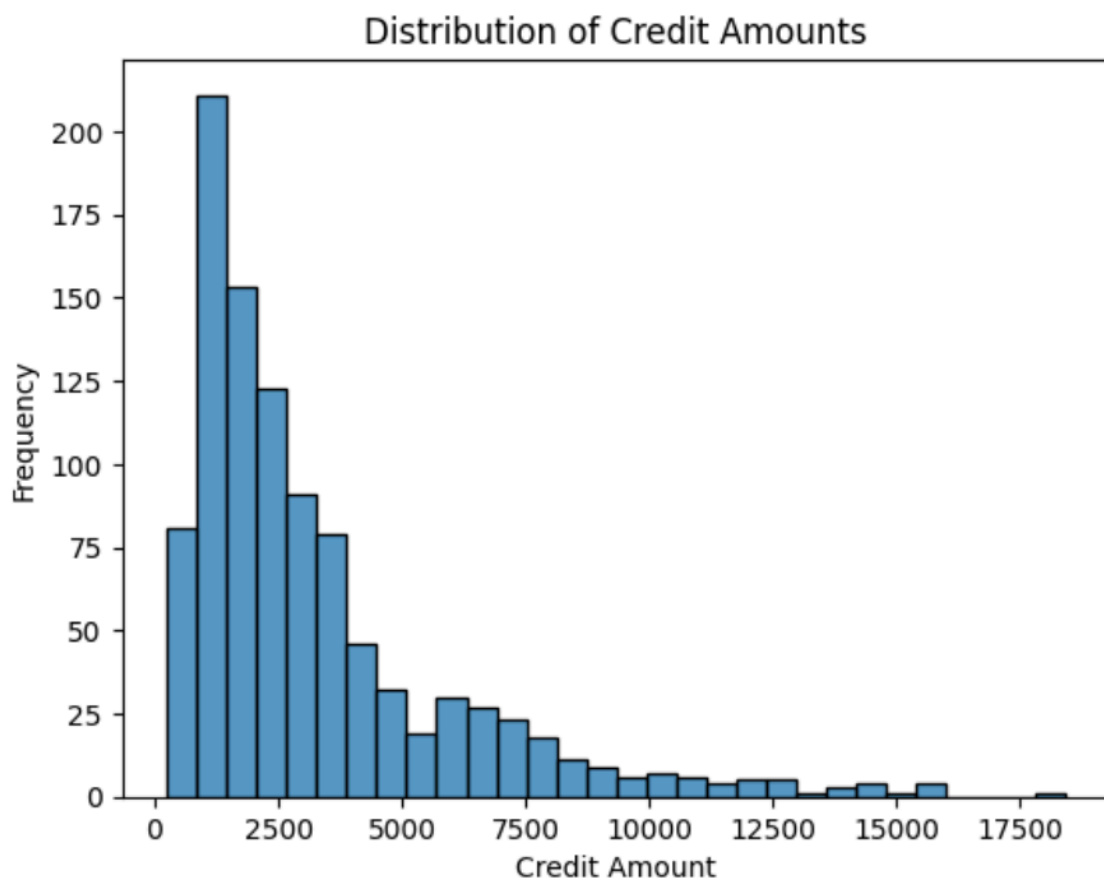
# Check structure
print(data.head())
print(data.info())
```

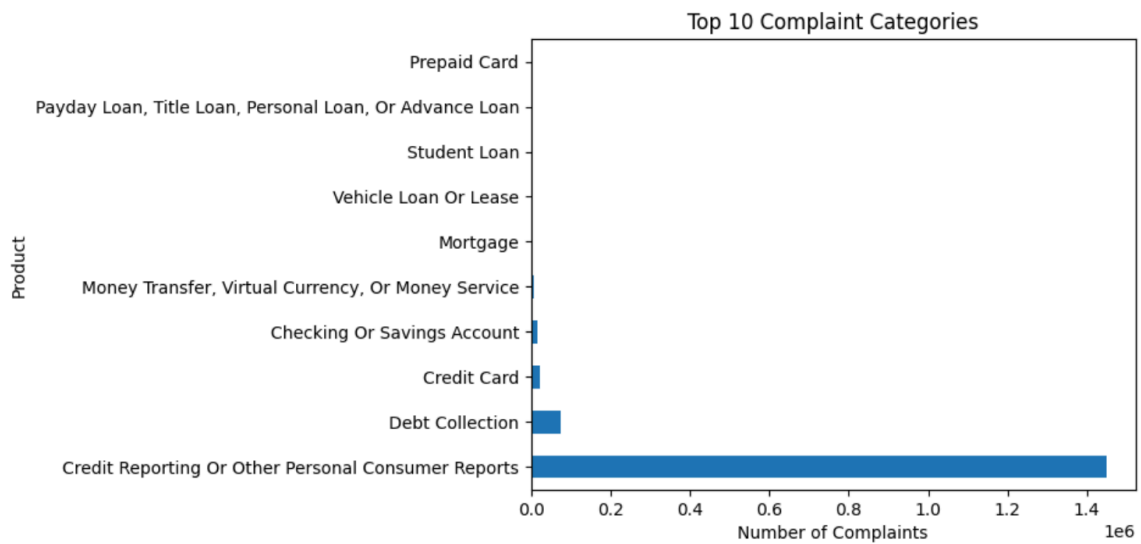
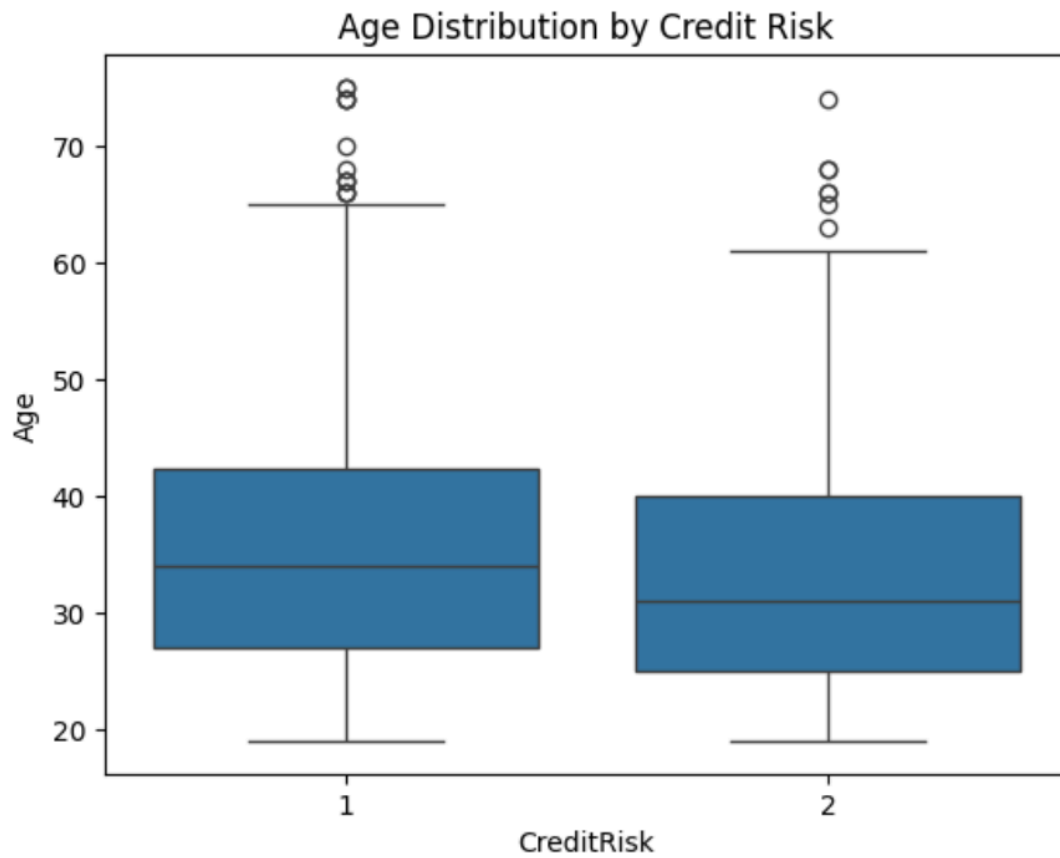
Group Number: 11269

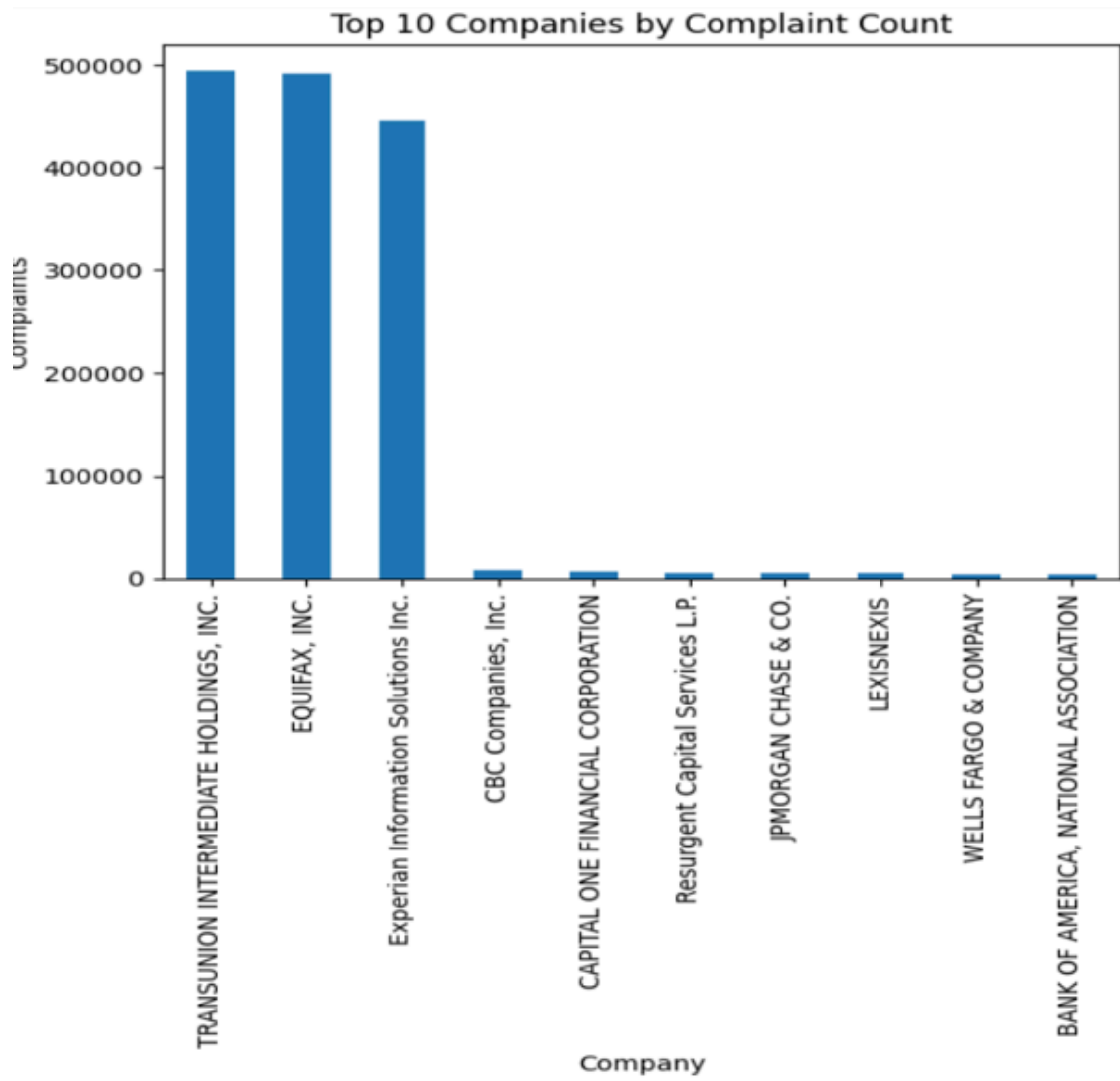
```
print(data['Class'].value_counts())
```

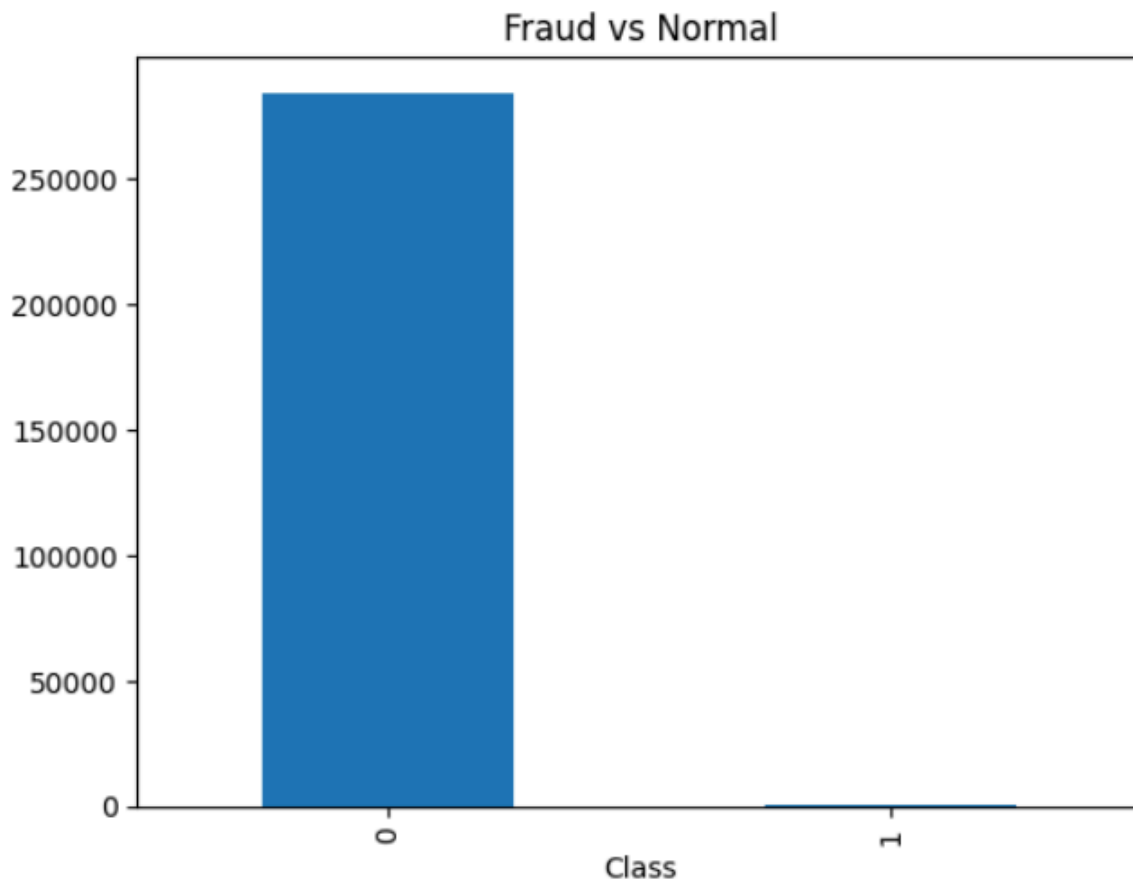
6. Exploratory Data Analysis of Sample Data

Our data analysis shows that for the first data set of German Credit Data loans are concentrated to smaller amounts as well as young people are at more risk for credit due to lower median in credit risk 2. For the second dataset of consumer complaints the credit complaints lead the number of complaints raised which in turn have a relationship to creditworthiness of individuals. The third dataset for Kaggle credit card fraud class 1 i.e. fraud-based transactions are on a very low side as compared to class 0 normal transactions.



Group Number: 11269

Group Number: 11269

Group Number: 11269

7. Short literature searches that links to papers citing research

Conventional credit assessment, used by banks and other lenders, primarily relies on traditional financial data to evaluate the repayment capacity of borrowers (Vanini et al. 2023). This conventional approach can unfortunately exclude creditworthy entities, particularly new ventures and start-ups, from accessing loans. By integrating alternative data—such as behavioral patterns, geolocation information, and digital application usage—a more nuanced evaluation becomes possible. This expanded data pool allows for credit models that are both more inclusive and precise, lowering the risk for lenders and offering a clearer view of the prevailing economic climate.

The predictive power of credit scoring is significantly strengthened by this supplementary data. Research indicates that incorporating details like transaction history, digital communication patterns, psychometric evaluations, and demographic information boosts the performance of established models. For example, one study demonstrated that models using digital footprints—including online purchase behavior, payment consistency, location data, and device information—outperformed traditional internal scoring systems by 18.4% in predicting default risk (Jiang et al. 2021). The reliability of these alternative methods is further supported by evidence that credit grades from platforms like Lending Club, which use such data, are on par with those issued by conventional banks (Grace 2020).

Group Number: 11269

Furthermore, non-traditional data serves as a critical tool for risk mitigation by helping to identify potential fraudulent behavior. Analysis of legal costs, for instance, can reveal dishonesty in loan applications and weaknesses in internal oversight (McNulty and Akhigbe 2017). The influence of social connections is also a significant factor; studies of peer-to-peer lending show that individuals within tight-knit networks often exhibit similar creditworthiness (Caglayan et al. 2021). This "contagion" effect extends to the workplace, where an analysis of US financial advisors revealed that employees are more likely to commit misconduct if they are teamed with new colleagues who have a prior history of wrongdoing, a dynamic often exposed during corporate mergers (Dimmock et al. 2018).

Beyond individual risk, this data provides a macro-economic lens for identifying market trends. Analysis of millions of auto loans, for example, shows that the length of a loan term directly impacts the prices of durable goods by shaping consumer demand. A one-year extension in loan maturity can lead to an approximate 2.8% increase in a car's price (Argyle et al. 2020).

In summary, leveraging alternative data sources transforms credit evaluation. It injects greater accuracy into lending decisions, deepens the understanding of borrower behavior, and provides critical insights into broader market dynamics, ultimately creating a more robust and equitable financial system.

Group Number: 11269

References

Consumer Complaint Database

<https://www.consumerfinance.gov/data-research/consumer-complaints/>

Accessed 17 Oct. 2025

Kaggle Credit Card Fraud

<https://www.kaggle.com/mlg-ulb/creditcardfraud/>

Accessed 17 Oct 2025

Liu, Grace. *Data Quality Problems Troubling Business and Financial Researchers:*

A Literature Review and Synthetic Analysis. Journal of Business & Finance

Librarianship, November 2020, https://digitalcommons.wcupa.edu/lib_facpub/13/.

Vanini P, Rossi S, Zvizdic E, Domenig T (2023) *Online payment fraud: from anomaly detection to risk management*. <https://jfin-swufe.springeropen.com/articles/10.1186/s40854-023-00470-w>

McNulty JE, Akhigbe A (2017) *What do a bank's legal expenses reveal about its internal controls and operational risk?* <https://doi.org/10.1016/j.jfs.2016.10.001>

Argyle B, Nadauld T, Palmer C, Pratt R (2020) *The capitalization of consumer financing into durable goods prices*. <https://onlinelibrary.wiley.com/doi/10.1111/jofi.12977>

Dimmock SG, Gerken WC, Graham NP (2018) *Is fraud contagious? coworker influence on misconduct by financial advisors* <https://onlinelibrary.wiley.com/doi/10.1111/jofi.12613>